EXPERIMENT NO: 21

DISPLAY A NUMBER IN SEVEN SEGMENT LED IN LPC2148 KIT

AIM

To write and execute C program to display a number in seven segment LED in LPC2148 kit.

OBJECTIVES

• Configure the LPC2148 GPIO pins for seven-segment segment data and display selection.

• Generate the required segment codes for displaying decimal digits 0 to 9.

• Continuously increment and display the count from 0 to 9 with a one-second delay.

APPARATUS / SOFTWARE REQUIRED

• Keil uVision5 Software

• Philips Flash Programmer

• LPC 2148 kit

PROGRAM

```
//SEVEN SEGMENT LED DISPLAY INTERFACE IN C
/* Program to Count 0-9 and Display it in 7 segment Display (MUX) DS4
 * Display Select DS3 ==> "P0.13" Enable --> '0', Disable --> '1'
 * Display Select DS4 ==> "P0.12" Enable --> '0', Disable --> '1'
 */
/* Segment Connection Display 1 & 2 Enable --> '1', Disable --> '0'
 *------------------------------------------------------------------
 * MSB LSB
 * Dp G F E D C B A
 * P0.23 P0.22 P0.21 P0.20 P0.19 P0.18 P0.17 P0.16
 *-------------------------------------------------------------------*/

#include <LPC214X.H>

#define DS3 1<<13
#define DS4 1<<12
#define SEG_CODE 0xFF<<16

unsigned char const seg_dat[]={0x3F, 0x6, 0x5B, 0x4F, 0x66, 0x6D, 0x7D, 0x7, 0x7F, 0x67};

void delayms(int n)
{
    int i,j;
    for(i=0;i<n;i++)
    {
        for(j=0;j<5035;j++) //5035 for 60Mhz ** 1007 for 12Mhz
        {
        }
    }
}

int main (void)
{
    unsigned char count;

    PINSEL0 = 0;
    PINSEL1 = 0;
    IODIR0 = SEG_CODE | DS3 | DS4;
    IOSET0 = SEG_CODE | DS3;
    IOCLR0 = DS4;
    count = 0;

    IOCLR0 = SEG_CODE;
    IOSET0 = seg_dat[count]<<16;

    while(1)
    {
        delayms(1000);
        count++;
```

```
            if(count>9) count=0;

            IOCLR0 = SEG_CODE;
            IOSET0 = seg_dat[count]<<16;
        }
    }
```

OUTPUT

7-Segment display counting from 0 to 9

RESULT

Thus C program was written and executed to display a number in seven segment LED in LPC2148 kit.